R FOR DATA ANALYSIS · CLEANING AND RECODING · LESSON 2

02 Group numbers into ranges with cut()

Turn household size into groups like 1 to 5 and 5 to 10, then count them with cross_cases().

BY THE END OF THIS SECTION YOU WILL BE ABLE TO

- Create ranges with cut(), breaks and labels
- Use Inf for an open top range
- Count each range with cross_cases()

The R code in this lesson

```
library(expss)

# Make ranges from a number column
demographic <- demographic %>% mutate(HH_Size = cut(HH_Size,
                      breaks = c(1, 5, 10, 15, 20, Inf),
                      labels = c("1 to 5", "5 to 10", "10 to 15",
                                 "15 to 20", "More than 20")))

# Frequency table
cross_cases(demographic, cell_vars = HH_Size)
```

What you will hear (times count from the end of the intro)

Intro A long list of numbers is hard to read. Groups are much easier: small, medium, and large families, for example. In this lesson, you will learn to group numbers into ranges with cut, and then count each group in a neat table. Let's make our numbers tell a story.

0:00 On the right, the Viewer shows a frequency table of household size. Every single size has its own row: three, four, five, seven, and so on. It is long and hard to read.

0:10 I write a note: to make a range of column in a dataset. Then demographic, pipe, mutate.

0:30 Inside mutate: H H Size equals cut. Cut takes the number column, and cuts it into pieces.

0:45 First, breaks. The breaks are the edges of each group: one, five, ten, fifteen, twenty, and Inf.

1:00 Inf means infinity. It makes the last group open at the top, so every big family is included, no matter how large.

1:15 Then labels, one for each group, in quotes: one to five, five to ten, ten to fifteen, fifteen to twenty, and more than twenty.

1:45 Be careful with the brackets. There are several levels here, so close each one, slowly.

2:40 Run it, and open the table. The household size column now shows the group names instead of numbers.

2:55 Now let's count them. Cross underscore cases, from the expss package, with cell vars equals H H Size.

3:12 And here is our new table. Twenty four households have one to five members, and fifty seven have more than twenty. Much clearer! Wonderful work.